Tian Yueyao

AI & Signal Processing Engineer | Human-Machine Interaction Researcher

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CONTACT

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PORTFOLIO

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EDUCATION

Master of Science
(2024 – 2026)

Signal Processing & Machine Learning
Tampere University, Finland | GPA: 3.81 / 5.0

- Core Focus:
  Advanced Signal Processing, Deep Learning, Audio & Speech Processing.
- Thesis: Chaotic Hierarchical Network for Speaker Recognition.

Bachelor of Engineering
(2020 – 2024)

Software Engineering
Tiangong University, China | GPA: 3.52 / 4.0

- Quantitative: Advanced Mathematics | Data Statistics based on R Language.
- Engineering: Algorithm Design | Data Structures | Oracle Database.
- Systems: Object-Oriented Programming (Java) | System Analysis.

EXPERIENCE

CNN-Based Automatic Music Transcription (Sept – Dec 2024)

- **Developed** a multi-label classification system for audio-to-MIDI conversion using the MAESTRO dataset.
- **Optimized** temporal dependency modeling for rhythmic onset detection via CNN architectures.
- **Technologies:** PyTorch, Librosa, STFT, Mel-spectrograms.

Robot Scientific Research | Team Leader (2021 – 2023)

- **Led** an interdisciplinary team to design real-time sensor fusion and obstacle avoidance algorithms.
- **Managed** high-frequency data synchronization across multiple hardware/software interfaces.
- **Impact:** Demonstrated robust project management in a complex, multi-modal engineering environment.

RESEARCH

Master's Thesis: Chaotic Speaker Recognition System (2025 – 2026)

Tampere University | Supervisor: Annamaria Mesaros

- **Developed** the ChaoNet framework for advanced audio analysis.
- **Modeled** complex acoustic events using non-linear dynamical systems (Lorenz, Rössler, Chua and Mackey-Glass).
- **Implemented** phase space reconstruction based on Takens' embedding theorem to capture deterministic structures in noisy signals.
- **Applied** RQA and MLSA analysis to quantify temporal dependencies in audio data.
- **Result:** Achieved 96.4% accuracy on the LibriSpeech dataset, demonstrating robust performance under noisy conditions.

Semantic Image Inpainting with GANs (Sept – Dec 2024)

Tampere University | Advanced Image Processing Course

- Designed a generative adversarial network (GAN) for high-fidelity image restoration.
- Optimized latent space via context and prior loss functions to ensure visual realism.
- Integrated Poisson blending to achieve seamless synthesis in large missing regions.

EXPERTIES

- **Programming:**
  Python · MATLAB · R · C· Java
- **Frameworks:**
  PyTorch · TensorFlow · Scikit-learn
- **Audio/Signal:**
  Time-frequency Analysis · Chaos Theory
  Biosignal Processing (ECG/PPG)
- **Tools:**
  Linux/SLURM · Git · LaTeX

LANGUAGE

English: Full professional proficiency

Finnish: Basic

Korean: Conversational

Soft Skills:
Interdisciplinary collaboration, complex problem solving, and adaptive learning in fast-paced research environments.